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Neostomy apparatus

Abstract

A neostomy apparatus, including a grasping device, a cutting device and a control handle. The control handle includes a housing and a manipulation portion disposed in the housing, the manipulation portion can move axially relative to the housing; the grasping device or the cutting device includes an elongated member, and the manipulation portion drives the elongated member to move axially; the elongated member is partially located in the housing, and a first limiting structure is further disposed in the housing, the first limiting structure has a groove or a through hole, and a portion of the elongated member located in the housing is at least partially disposed in the groove or the through hole of the first limiting structure.

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Background/Summary

TECHNICAL FIELD

(1) The embodiments relate to the field of medical apparatuses, in particular to a neostomy apparatus.

BACKGROUND

(2) Heart failure (referred to as “cardiac failure”) is a group of complex clinical syndromes caused by ventricular filling or impaired ejection function due to any heart structural or functional abnormality, which is mainly manifested as dyspnea and asthenia (limited exercise tolerance), and fluid retention (pulmonary congestion and peripheral edema) clinically. Heart failure, as a severe stage developed from various heart diseases, is becoming the most important cardiovascular disease throughout the world.

(3) Heart failure can be divided into diastolic heart failure and systolic heart failure. Diastolic heart failure refers to a congestion syndrome of pulmonary circulation and systemic circulation caused by the decrease of ventricular filling volume and the increase of filling pressure due to the decrease of ventricular relaxation and compliance in case of normal ventricular contraction functions. Diastolic heart failure will result in increased pressure in the left atrium and pulmonary veins, hindering the normal flow of oxygen bearing blood.

(4) Currently, there are a few of methods for treating diastolic heart failure patients. Clinical data show that a small hole is opened on the interatrial septum of a diastolic heart failure patient to form a left-to-right shunt, which is beneficial to reducing the pressure of the heart failure patient in the left atrium. To open a small hole on interatrial septum requires a simple operation of the neostomy apparatus; especially the handle accords with the ergonomics and the control mechanism on the handle is simple.

SUMMARY

(5) The embodiments provide a neostomy apparatus which is simple in construction and manipulation.

(6) The embodiments provide a neostomy apparatus, including a grasping device, a cutting device and a control handle; where, the control handle includes a housing and a manipulation portion disposed in the housing, the manipulation portion may axially move relative to the housing; the grasping device or the cutting device includes an elongated member, and the manipulation portion

drives the elongated member to move axially; the elongated member is partially located in the housing, and a first limiting structure is further disposed in the housing, the first limiting structure has a groove or a through hole, and a portion of the elongated member located in the housing is at least partially disposed in the groove or the through hole of the first limiting structure.

(7) In an embodiment, a proximal end of the first limiting structure is close to a distal end of the manipulation portion.

(8) In an embodiment, the grasping device further includes a catheter body, where the elongated member is disposed to pass through the catheter body, and a proximal end of the catheter body is connected to the housing; and the housing is provided with a sealing assembly; the sealing assembly is located at the distal end of the housing and is sealingly connected to the proximal end of the catheter body, and the catheter body may be communicated with a negative pressure source through the sealing assembly, such that a negative pressure is formed in the catheter body under the action of the negative pressure source.

(9) In an embodiment, the sealing assembly includes a hollow pipe body and a sealing member connected at both ends of the hollow pipe body; a vent hole is opened on a sidewall of the hollow pipe body, and the negative pressure source is communicated with the vent hole via a hose.

(10) In an embodiment, the manipulation portion is connected with a second limiting structure; the second limiting structure may house the elongated member and is coaxial with the first limiting structure; a proximal end of the second limiting structure and a proximal end of the elongated member are fixed on the manipulation portion; a distal end of the second limiting structure is inserted into the groove or the through hole of the first limiting structure and may move axially relative to the first limiting structure under the action of the manipulation portion.

(11) In an embodiment, a linear guide rail is fixedly disposed in the housing; the manipulation portion is slidably disposed on the linear guide rail; one or more arc-shaped grooves are disposed on a side of the linear guide rail; an elastic member and a ball are disposed in the manipulation portion; the ball is elastically abutted against one side of the linear guide rail opened with the arc-shaped groove under the action of the elastic member; and when the manipulation portion moves until the ball is opposite to the arc-shaped groove along the linear guide rail, partial structures of the ball are abutted into the arc-shaped groove.

(12) In an embodiment, the neostomy apparatus further includes a conductive structure and a conductive interface disposed at a proximal end portion of the housing; the conductive structure always keeps electric connection with the manipulation portion and the conductive interface when the manipulation portion moves axially.

(13) In an embodiment, the housing is provided with a conductive sliding rail, and a proximal end of the conductive sliding rail is connected to the conductive structure; and the manipulation portion is slidably connected to the conductive sliding rail, and the manipulation portion may move axially relative to the conductive sliding rail.

(14) In an embodiment, the conductive structure includes a conductive elastic piece, and the conductive elastic piece is disposed at a sliding connection portion between the manipulation portion and the conductive sliding rail; when the manipulation portion moves axially along the conductive sliding rail, the conductive elastic piece is always abutted against the conductive sliding rail.

(15) In an embodiment, the manipulation portion includes a sliding block and a pushing member; the sliding block is slidably disposed in the housing and is connected to the proximal end of the elongated member; the pushing member is connected to the sliding block, and partial structures of the pushing member are exposed to the housing, and the pushing member may move axially relative to the housing, such that the sliding block drives the elongated member to move axially.

(16) The embodiments provide a neostomy apparatus, including a control handle; the control handle adsorbs a tissue at a position which requires neostomy by negative pressure generated by the catheter body in communication with a negative pressure source. Moreover, a first manipulation

portion is utilized to drive a push rod to move axially in the catheter body such that a grasping portion connected to a distal end of the push rod is manipulated to grasp the tissue sucked into the catheter body, thereby facilitating the subsequent cutting and neostomy conducted on the tissue sucked into the catheter body by a second manipulation portion and a third manipulation portion. Moreover, buckle-proof or anti-buckling cannulas are sleeved on a push rod, a traction fiber and a control fiber of the cutting device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram illustrating a structure of a neostomy apparatus according to an embodiment of the embodiments;
- (2) FIG. 2 is a partially enlarged schematic diagram illustrating an internal structure of the neostomy apparatus shown in FIG. 1;
- (3) FIG. 3 is a schematic diagram illustrating an internal structure of a control handle of the neostomy apparatus according to an embodiment;
- (4) FIG. 4 is a partial schematic diagram illustrating an internal structure of the control handle of the neostomy apparatus shown in FIG. 3;
- (5) FIG. 5 is an exploded diagram of the control handle of the neostomy apparatus shown in FIG. 3;
- (6) FIG. 6 is a schematic diagram illustrating a structure of a first sliding block of the control handle according to an embodiment;
- (7) FIG. 7 is a schematic diagram illustrating a structure of a second sliding block of the control handle according to an embodiment;
- (8) FIG. 8 is a schematic diagram illustrating a cooperation structure of the first sliding block and a linear guide rail in the control handle according to an embodiment; and
- (9) FIG. 9 is a schematic diagram illustrating a structure of a third sliding block and a conductive elastic piece in the control handle according to an embodiment.

DETAILED DESCRIPTION OF THE EMBODIMENTS

- (10) For the convenience of understanding the embodiments, the embodiments will be described more comprehensively below with reference to relevant drawings.
- (11) Implementations of the embodiments are set forth in the drawings. However, the embodiments may be implemented in many different forms, but are not limited to the embodiments described herein. On the contrary, these embodiments are provided such that the embodiments are understood more thoroughly and comprehensively.
- (12) It can be understood that the terms “distal end” and “proximal end” are used as nouns of locality which are customary terms in the field of medical apparatus. The “distal end” denotes an end away from the operator during operation, and “proximal end” denotes an end close to the operator during operation. The axial direction refers to a direction parallel to the line of centers between the distal center and the proximal center of the medical apparatus; the radial direction refers to a direction perpendicular to the above axial direction.
- (13) Referring to FIGS. 1 and 2, the neostomy apparatus **100** provided by the embodiments includes a grasping device, a cutting device **20** and a control handle **30**. Both the grasping device and the cutting device include an elongated member partially located within the control handle. The elongated member may include a push rod, a traction firer, and a control firer.
- (14) The grasping device includes a catheter body **11**, a grasping portion **12** and a push rod **13**. A proximal end of the catheter body **11** is connected to the control handle **30**; the grasping portion **12** is disposed in the catheter body **11** and located at the distal end of the catheter body **11**; a distal end portion of the push rod **13** is sleeved on the grasping portion **12**, and the push rod **13** may move

axially in the catheter body **11**, so as to control the grasping portion **12** to perform a grasping operation.

(15) The catheter body **11** has a hollow tubular structure which may be communicated with a negative pressure source to form negative pressure in the catheter body **11** under the action of the negative pressure source. Thus, a tissue of interatrial septum is absorbed by the negative pressure formed in the catheter body **11**. The grasping portion **12** at the distal end of the catheter body **11** may grasp the tissue of human interatrial septum sucked into the lumen of the catheter body **11** by the neostomy apparatus **100** when it moves axially on the push rod **13**.

(16) The grasping portion **12** includes at least two oppositely disposed clamping jaws **121** and **122** which are fixed inside the distal end of the catheter body **11**. A distal end portion of the push rod **13** is sleeved on the grasping portion **12**, and thus, the push rod **13** may manipulate the opening or closure of the clamping jaws **121** and **122** in the grasping portion **12** when the push rod **13** moves axially relative to the catheter body **11**, thus achieving a clamping action.

(17) The cutting device **20** includes a cutting portion **21**, a traction fiber **22** and a control fiber **23**. The cutting portion **21** is disposed at the distal end of the catheter body **11** and partially surrounds the outer wall at an opening portion of the distal end of the catheter body **11**. The cutting portion **21** is configured for cutting the tissue of human interatrial septum sucked into the lumen of the catheter body **11** by the neostomy apparatus **100**. Both the distal ends of the traction fiber **22** and the control fiber **23** are connected to the cutting portion **21**.

(18) A first limiting structure, a push rod, a traction fiber and a control fiber are all partially located in the housing of the handle, and the portions located in the housing are all at least partially disposed in the groove or through hole of the first limiting structure. The structure is described in detail below.

(19) A control handle **30** is used for manipulating the grasping device **10** of the neostomy apparatus **100** for gripping and is further used for the cutting device **20** of the neostomy apparatus **100** for cutting.

(20) In combination with FIGS. **1** and **3**, the control handle **30** includes a housing **31** and a manipulation portion disposed within the housing **31**, and the manipulation portion manipulates the elongated member to move axially. The manipulation portion may particularly include a first manipulation portion **32**, a second manipulation portion **33** and a third manipulation portion **34**. The manipulation portion includes a sliding block and a pushing member; the sliding block is slidably disposed in the housing and connected to the proximal end of the elongated member; the pushing member is connected to the sliding block, and partial structures of the pushing member are exposed to the housing, and the pushing member may move axially relative to the housing, such that the sliding block drives the elongated member to move axially. In addition, an opening is opened on a sidewall of the housing, and a portion of the pushing member is exposed to the housing through the opening, and the opening has a certain length along the axial direction of the housing, and the axial length of the opening limits the axial moving range of the pushing member.

(21) The proximal end of the catheter body **11** is connected to the housing **31**, and a sealing assembly **350** is disposed in the housing **31**; the sealing assembly **350** is located at the distal end of the housing **31** and is sealingly connected to the proximal end of the catheter body **11**, such that after the catheter body **11** is in communication with the negative pressure source, the catheter body **11** may generate a negative pressure better under the action of the negative pressure source, so as to effectively suck the tissue of human interatrial septum into the lumen of the catheter body **11** for the subsequent grasping operation of the grasping device and the cutting operation of the cutting device **20**.

(22) As shown in FIG. **5**, the sealing assembly **350** includes a hollow pipe body **351** and a sealing member **352** connected at two ends of the hollow pipe body **351**; a vent hole **351a** is opened on a sidewall of the hollow pipe body **351**; and the negative pressure source is communicated with the vent hole **351a** via a hose **306**. Partial sections of the hose **306** may be disposed in the housing **31**

and may be connected to an external negative pressure source via a three-way valve **307** after being led out from the housing **31**, such that the three-way valve **307** may regulate the negative pressure source to act on the catheter body **11**, and accordingly, the catheter body **11** may adsorb the tissue of human interatrial septum with a suitable negative pressure.

(23) With continuing reference to FIG. 5, the housing **31** includes a first housing **311** and a second housing **312** which may be fastened together by a buckle, a screw or a glue bonding way. The distal end of the housing **31** may be tightly locked by a head cover **304**. For example, when the first housing **311** and the second housing **312** are buckled together, the distal end thereof forms an external thread, and the head cover **304** is in threaded fit connection to the distal end of the housing **31**, thereby locking the first housing **311** and the second housing **312** together and preventing the first housing **311** and the second housing **312** from loosening.

(24) In some embodiments, the proximal end of the housing **31** may enclose a connection port of a radio frequency probe in the form of a trailing housing **305**, such that the neostomy apparatus **100** is connected to an external high frequency equipment through the port. The neostomy apparatus **100** and a negative plate on the body surface form a current loop in operation, thereby satisfying the power-on needs of the cutting device **20** for operation.

(25) It may be understood that the proximal end of the housing **31** refers to the end where the housing **31** faces the operator during handling and using, after being assembled to the neostomy apparatus.

(26) With continuing reference to FIGS. 1 and 3, the housing **31** may be divided into a distal end section **301**, a proximal end section **302**, and a body section **303** connected between the distal end section **301** and the proximal end section **302** when the control handle **30** is manipulated. The distal section **301** may serve as a holding area for stabilizing the neostomy apparatus **100** during operation. An arc-shaped convex structure may be formed on an upper surface of the proximal section **302** to press against the palm to manipulate the control handle **30**. For example, a first manipulation portion **32**, a second manipulation portion **33** and a third manipulation portion **34** are distributed on the body section **303**, used for manipulating the grasping device **10** and the cutting device **20**, respectively. In other embodiments, the distribution position, i. e., the order, of the first manipulation portion **32**, the second manipulation portion **33**, and the third manipulation portion **34** is not limited in the housing **31** as long as these portions are adapted to the corresponding manipulation requirements only.

(27) The first manipulation portion **32**, the second manipulation portion **33** and the third manipulation portion **34** are respectively used for manipulating the axial movement of the push rod **13**, the traction fiber **22** and the control fiber **23** in the catheter body **11**, respectively, as an example, to further describe the structure of the manipulation handle.

(28) For example, in combination with FIGS. 3 to 5, the first manipulation portion **32** is connected to the proximal end **13a** of the push rod **13** and is configured for driving the push rod **13** to move axially relative to the catheter body **11**, such that the grasping portion **12** opens or closes under the axial movement of the push rod **13**, thereby achieving the grasping manipulation of the grasping portion **12** via the first manipulation portion **32**.

(29) The second manipulation portion **33** is connected to the proximal end **22a** of the traction fiber **22** and is configured for driving the traction fiber **22** to move axially in the catheter body **11**. Since the traction fiber **22** is connected to the cutting portion **21** of the cutting device **20**, the second manipulation portion **33** drives the traction fiber **22** to move axially in the catheter body **11**, such that the cutting portion **21** may be suspended at the distal end of the catheter body **11** transformed from a tight state bound to the catheter body **11**. Accordingly, the third manipulation portion **34** is connected with the proximal end **23a** of the control fiber **23** and is configured for driving the control fiber **23** to move axially in the catheter body **11**, thus holding the control fiber **23** into the catheter body **11**.

(30) In some embodiments, the traction fiber **22** and the control fiber **23** are symmetrically

disposed on the wall of the catheter body **11** to exert an axial push-pull force on the opposite sides of the cutting portion **21** connected on the distal ends of the traction fiber **22** and the control fiber **23**. In other embodiments, the traction fiber **22** and the control fiber **23** may also be disposed asymmetrically as long as it is ensured that the cutting portion **21** may be manipulated by the axial movement of the traction fiber **22** and the control fiber **23** in the catheter body **11**. Therefore, the cutting portion **21** cuts the tissue of human interatrial septum sucked into the catheter body **11**.

(31) In some embodiments, the first limiting structure includes a cannula. For example, a first sleeve pipe **361** is fixedly disposed between the sealing assembly **350** and the first manipulation portion **32** in the housing **31**; and the push rod **13** may be movably disposed to pass through the first sleeve pipe **361** axially, such that when the first manipulation portion **32** drives the push rod **13** to move towards the distal end due to the guiding effect of the first sleeve pipe **361** on the push rod **13**, the push rod **13** is not easily bent due to the radial constraint of the first sleeve pipe **361**, thereby improving the manipulation reliability.

(32) It may be understood that the first sleeve pipe **361** serves as a limiting structure to produce a restraining effect in the radial direction when the push rod **13** moves in the axial direction, and the structure is configured to adapt to the first manipulation portion **32** such that the push rod **13** is driven to move axially, thereby manipulating the grasping portion **12** for grasping operation. Further, the configuration of the first sleeve pipe **361** does not interfere with the first manipulation portion **32** to manipulate the push rod **13** to move axially. For example, in some embodiments, the proximal end surface of the first sleeve pipe **361** just contacts the side on which the distal end of the first manipulating portion **32** is located when the push rod **13** is manipulated by the first manipulation portion **32** to move axially toward the distal end to a limiting position. In another example, in other embodiments, there is an interval or gap between the proximal end surface of the first sleeve pipe **361** and the side on which the distal end of the first manipulation portion **32** is located when the push rod **13** is manipulated by the first manipulation portion **32** to move axially toward the distal end to a limiting position. Whether the proximal end surface of the first sleeve pipe **361** is just in contact with or has a gap with the side on which the distal end of the first manipulation portion **32** is located, it is ensured that the first sleeve pipe **361** will not interfere with the movement of the first manipulation portion **32** towards the distal end when the push rod **13** is driven by the first manipulation portion **32** to manipulate the grasping portion **12** for operation.

(33) With continuing reference to FIGS. **3** and **5**, in some embodiments, a second sleeve pipe **362** and a third sleeve pipe **363** are fixedly disposed in the housing **31**; distal ends of the second sleeve pipe **362** and the third sleeve pipe **363** are respectively fixed to the proximal end of the sealing assembly **350**. The traction fiber **22** may be axially disposed to pass through the second sleeve pipe **362** movably, and the control fiber **23** may be axially disposed to pass through the third sleeve pipe **363** movably, such that the second sleeve pipe **362** and the third sleeve pipe **363** may have a better radial restraining effect on the traction fiber **22** and the control fiber **23**, respectively. Thereby, the above configuration prevents the traction fiber **22** and the control fiber **23** from being easily bent under the pushing force of the second manipulation portion **33** and the third manipulation portion **34**.

(34) Two or more spaced supporting sheets **308** are disposed in the housing **31** at a position near the distal end; the supporting sheets **308** are fixed on the inner surface of the housing **31**. Ports **308a** are opened on the setting paths of the first sleeve pipe **361**, the second sleeve pipe **362** and the third sleeve pipe **363**, such that the first sleeve pipe **361**, the second sleeve pipe **362** and the third sleeve pipe **363** are located in the corresponding ports **308a**, thus stably fixing the first sleeve pipe **361**, the second sleeve pipe **362** and the third sleeve pipe **363** in the housing **31** by the ports **308a**.

(35) The ports **308a** may be a circular arc groove having a diameter comparable to or slightly smaller than that of the first sleeve pipe **361**, the second sleeve pipe **362** and the third sleeve pipe **363**.

(36) A gland **309** may be disposed between the supporting sheets **308** to further fasten the first

sleeve pipe **361**, the second sleeve pipe **362**, and third sleeve pipe **363**.

(37) For example, both ends of the gland **309** are respectively fixed to the housing **31** by buckling, welding, glue bonding or screws, or the like. The gland **309** is pressed on the first sleeve pipe **361**, the second sleeve pipe **362** and the third sleeve pipe **363**, so as to achieve a better stabilizing effect and prevent the first sleeve pipe **361**, the second sleeve pipe **362** and the third sleeve pipe **363** from loosening in the housing **31**, thus affecting the guiding effect.

(38) The diameters of the first sleeve pipe **361**, the second sleeve pipe **362** and the third sleeve pipe **363** may or may not be the same so long as they are adapted to produce a radial restraining effect on the inner members thereof, thus achieving the purpose of preventing buckling. Taking the first sleeve pipe **361** as an example, the bore diameter of the first sleeve pipe **361** may be slightly greater than the outer diameter of the push rod **13**, such that when the push rod **13** is subjected to an axial pressing force and thus has a tendency of radial deflection, the first sleeve pipe **361** with a slightly bore diameter may prevent the radial deflection of the push rod **13**, thereby achieving a better anti-buckling effect. It can be appreciated that, in other embodiments, the cannula may be disposed alternatively; for example, a cannula may be disposed only outside the control fiber and the traction fiber, but not disposed outside the push rod.

(39) In some embodiments, the manipulation portion is further provided with a second limiting structure, the second limiting structure includes a telescopic sleeve. For example, a first telescopic sleeve pipe **371** is connected on the second manipulation portion **33**; the first telescopic sleeve pipe **371** is coaxially sleeved between the traction fiber **22** and the second sleeve pipe; the proximal end of the first telescopic sleeve pipe **371** and the proximal end **22a** of the traction fiber **22** are fixed to the second manipulation portion **33**; the distal end of the first telescopic sleeve pipe **371** is inserted into the second sleeve pipe **362** and may move axially relative to the second sleeve pipe **362** under the action of the second manipulation portion **33**. And during the movement of the second manipulation portion **33**, the distal end of the first telescopic sleeve pipe **371** may not be exposed from the proximal end of the second sleeve pipe **362**. In this embodiment, the first telescopic sleeve pipe **371** may be telescopically positioned between the traction fiber **22** and the second sleeve pipe **362** to further enhance the radial constraint on the traction fiber **22**, thus effectively resisting buckling and maintaining the ability of the traction fiber **22** to move axially relative to the second sleeve pipe **362**. In this configuration, the proximal end of the first telescopic sleeve pipe **371** and the proximal end of the traction fiber **22** are fixed on the second manipulation portion **33**; and the distal end of the first telescopic sleeve pipe **371** is inserted into the second sleeve pipe **362**.

Thereby, when the second manipulation portion **33** drives the traction fiber **22** to move axially, the traction fiber **22** always gains the restraining effect of the first telescopic tube **371** in the radial direction. Therefore, even when the proximal end **22a** of the traction fiber **22** stretches out a relatively long distance from the proximal end **362a** of the second sleeve pipe **362**, the traction fiber **22** is not easily radially deflected due to the radial constraint of the first telescopic sleeve pipe **371**. Thereby, the traction fiber **22** may maintain a better axial movement performance under the axial telescopic movement between the first telescopic sleeve pipe **371** and the second sleeve pipe **362**, thus enhancing the manipulation effect of the second manipulation portion **33** on the traction fiber **22**.

(40) In some embodiments, a second telescopic sleeve pipe **372** is connected on the third manipulation portion **34**; the second telescopic sleeve pipe **372** is coaxially sleeved between the control fiber **23** and the third sleeve pipe. The proximal end of the second telescopic sleeve pipe **372** and the proximal end **23a** of the control fiber **23** are fixed to the third manipulation portion **34**, and the distal end of the second telescopic sleeve pipe **372** is inserted into the third sleeve pipe **363** and may move axially relative to the third sleeve pipe **363** under the action of the third manipulation portion **34**. During the movement of the third manipulation portion **34**, the distal end of the second telescopic sleeve pipe **372** is not exposed from the proximal end of the third sleeve pipe **363**. The second telescopic sleeve pipe **372** is axially and telescopically inserted into the third

sleeve pipe **363**, thus being meeting the anti-buckling requirement when the control fiber **23** is pushed and pulled with a large stroke in an axial direction. In particular, the proximal end of the second telescopic sleeve pipe **372** is fixed to the third manipulation portion **34** together with the proximal end **23a** of the control fiber **23**, such that when the second manipulation portion **33** drives the control fiber **23** to move axially, the second telescopic sleeve pipe **372** always exerts a better radial restraining effect on the control fiber **23** to prevent the control fiber **23** from buckling.

(41) It can be further understood that the diameter and length of the first and second telescopic sleeves **371** and **372** are adapted to the radial constraint requirements of the internal components thereof only. Taking the second telescopic sleeve pipe **372** as an example, the second telescopic sleeve pipe **372** is sleeved between the control fiber **23** and the third sleeve pipe **363**, and the inner wall of the second telescopic sleeve pipe **372** may be closely coated on the control fiber **23**, and may also form a certain gap with the outer wall of the control fiber **23**. Accordingly, the inner wall of the third sleeve pipe **363** may be closely coated on the second telescopic sleeve pipe **372**, and may also form a certain gap with the outer wall of the second telescopic sleeve pipe **372**. The above configuration may not only maintain the smooth movement of the control fiber **23** and the second telescopic sleeve pipe **372** relative to the third sleeve pipe **363** in the axial direction, but also may limit the radial deflection of the control fiber **23** by the axial telescopic movement of the second telescopic tube **372** in the third sleeve pipe **363**, thereby effectively ensuring that the control fiber **23** is free of buckling, and improving the manipulation reliability.

(42) It can be appreciated that in this embodiment, the axial displacement distance of the first manipulation portion is smaller than that of the second manipulation portion and the third manipulation portion. Moreover, the outer diameter of the push rod is larger than that of the control fiber and the traction fiber. Therefore, other telescopic sleeves may be not disposed between the first sleeve pipe and the control fiber. That is to say, whether a sleeve pipe or a telescopic sleeve pipe is disposed according to the displacement distance and the outer diameter of the corresponding fiber or rod. A person skilled in the art may make corresponding changes, substitutions or choices according to actual situations.

(43) It can be understood that in other embodiments, the limiting structure may also be a guide rail fixed on the inner wall of the housing, and the guide rail is provided with a groove or a through hole; and the elongated member is disposed in the groove or the through hole; alternatively, the limiting structure may be a part of the pipe body; the limiting structure forms a groove. That is, the cross section of the limiting structure is a section of arc, such that the elongated member is received within the groove.

(44) In some embodiments, in combination with FIGS. **4** and **5**, the first manipulation portion **32** includes a first sliding block **321** and a first pushing member **322**. The first sliding block **321** is slidably disposed in the housing **31** and is connected to the proximal end **13a** of the push rod **13**; the first pushing member **322** is connected to the first sliding block **321**, and partial structures of the first pushing member **322** are exposed to the housing **31**; and the first pushing member **322** may move axially relative to the housing **31**, such that the first sliding block **321** drives the push rod **13** to move axially.

(45) In combination with FIG. **6**, in some embodiments, the first sliding block **321** may be rectangular. In other embodiments, the first sliding block **321** may also be in other shapes, for example, the first sliding block **321** may be in a triangular or elliptical shape on a cross section perpendicular to the axial direction; the shape of the first sliding block **321** is not defined herein as long as the first sliding block **321** moves along the linear guide rail **380**, thus driving the push rod **13** to move axially in the catheter body **11**.

(46) The first sliding block **321** is provided with one or more first connecting holes **3211** such that the first connecting hole **3211** is matched with connecting members, such as, a screw or a bolt, thus fixedly connecting the first pushing member **322** with the first sliding block **321**.

(47) The first sliding block **321** is provided with a first insertion hole **3212** extending in an axial

direction, and a positioning hole **3213** communicating through the first insertion hole **3212** from a radial direction; the proximal end **13a** of the push rod **13** is inserted into the first insertion hole **3212**; and locking members, such as, a screw or a top thread, are fastened to the positioning hole **3213** and abutted against the proximal end **13a** of the push rod **13** in the first insertion hole **3212**, such that the proximal end **13a** of the push rod **13** is fixedly connected on the first sliding block **321**.

(48) The second manipulation portion **33** and the third manipulation portion **34** are both located on one side where the proximal end of the first manipulation portion **32** is located; the first sliding block **321** is provided with a first through hole **3213** and a second through hole **3214** which extend in an axial direction. The second sleeve pipe **362** and the third sleeve pipe **363** are axially and movably disposed to pass through the first through hole **3213** and the second through hole **3214**, respectively. Such a configuration mode achieves that the second sleeve pipe **362** is utilized to exert an anti-buckling effect on the traction fiber **22** and the third sleeve pipe **363** is utilized to exert an anti-buckling effect on the control fiber **23**; and the second sleeve pipe **362** and the third sleeve pipe **363** may be axially and movably disposed to pass through the first sliding block **321** of the first manipulation portion **32**, not interfering with the axial movement of the first sliding block **321**.

(49) It can be appreciated that the second sleeve pipe **362** and the third sleeve pipe **363** may be disposed axially and movably to pass through the first sliding block **321** of the first manipulating portion **32**, such that when the push rod **13** is driven by the first sliding block **321** to move axially in the catheter body **11**, the second sleeve pipe **362** and the third sleeve pipe **363** may always be supported by the first sliding block **321**, thus obtaining axial stability. Accordingly, the second sleeve pipe **362** may have a better guiding effect on the traction fiber **22** in the axial direction so as to prevent the traction fiber **22** from deviating radially. Correspondingly, the third sleeve pipe **363** may have a better guiding effect on the control fiber **23** in the axial direction so as to prevent the control fiber **23** from deviating radially.

(50) The second manipulating portion **33** and the third manipulating portion **34** are sequentially disposed in the axial direction on the side where the proximal end of the first manipulating portion **32** is located.

(51) In this embodiment, the second manipulation portion **33** includes a second sliding block **331** and a second pushing member **332**; the second sliding block **331** is slidably disposed in the housing **31** and is connected to the proximal end **22a** of the traction fiber **22**; the second pushing member **332** is connected to the second sliding block **331**, and partial structures of the second pushing member **332** are exposed to the housing **31**, and the second pushing member **332** may move axially relative to the housing **31**, such that the traction fiber **22** is driven by the second sliding block **331** to move axially.

(52) In combination with FIGS. **4** and **5**, a linear guide rail **380** is fixedly disposed in the housing **31**, and the first sliding block **321** and the second sliding block **331** are slidably disposed on the linear guide rail **380**. For example, the first sliding block **321** and the second sliding block **331** are both provided with a sliding groove or a sliding hole **30a** in sliding fit with the linear guide rail **380**, such that the linear guide rail **380** is slidably disposed to pass through the sliding groove or the sliding hole **30a**. The above configuration may achieve the sliding fit of the first sliding block **321** and the second sliding block **331** with the linear guide rail **380**. Understandably, since the first sliding block **321** is required to drive the push rod **13** to move axially and the second sliding block **331** is required to drive the traction fiber **22** to move axially, the linear guide rail **380** is substantially axially disposed in the housing **31**.

(53) The sliding groove or sliding hole **30a** may be rectangular or triangular in the cross section perpendicular to an axial direction, or "I"-shaped. Accordingly, the linear guide rail **380** has a structural form adapted to the sliding groove or sliding hole **30a**, which will not be described in detail herein.

(54) The second sliding block **331** has a structure similar to that of the first sliding block **321**. For

example, in combination with FIG. 7, in some embodiments, the second sliding block **331** may be rectangular. In other embodiments, the second sliding block **331** may also be in other shapes. For example, the second sliding block **331** may be in a triangular or elliptical shape on the cross section perpendicular to an axial direction. The shape of the second sliding block **331** is not defined here provided that the second sliding block **331** may move along the linear guide rail **380** to drive the traction fiber **22** to move axially in the catheter body **11**.

(55) The second sliding block **331** is provided with one or more second connecting holes **3311**, such that the second connecting hole **3311** is matched with connecting members, such as, a screw or a bolt, thus fixedly connecting the second pushing member **332** with the second sliding block **331**. Accordingly, the second sliding block **331** is provided with a second insertion hole **3312** extending in an axial direction, and a locking hole **3313** penetrating and communicating with the second insertion hole **3312** in a radial direction. The proximal end **22a** of the traction fiber **22** and the proximal end of the first telescopic sleeve pipe **371** are inserted into the second insertion hole **3312**; locking members, such as, a screw or a top thread, are fastened to the locking hole **3313** and abutted against the traction fiber **22** and the first telescopic sleeve pipe **371** in the second insertion hole **3312**, such that the proximal end **22a** of the traction fiber **22** and the proximal end of the first telescopic sleeve pipe **371** are fixedly connected on the second sliding block **331**.

(56) It can be understood that in some embodiments, the first telescopic sleeve pipe **371** outside the traction fiber **22** may be omitted, i. e., the first telescopic sleeve pipe **371** need not be connected on the second sliding block **331**, such that the traction fiber **22** may be disposed directly to pass through the second sleeve pipe **362**. And in this case, the locking member may be engaged with the locking hole **3313** in threaded fit and abutted against the proximal end **22a** of the traction fiber **22**, such that the second sliding block **331** manipulates the traction fiber **22** to move axially in the catheter body **11** under the action of the second pushing member **332**.

(57) The second sliding block **331** is provided with a third through hole **3314**, and the third sleeve pipe **363** may be axially disposed to pass through the third through hole **3314** movably, such that the control fiber **23** disposed to pass through the third sleeve pipe **363** may pass through the second sliding block **331**, and the proximal end **23a** of the control fiber **23** is connected to the third manipulation portion **34**. Further, the second sliding block **331** may move axially relative to the third sleeve pipe **363**, such that the second sliding block **331** is not interfered by the third sleeve pipe **363** when driving the traction fiber **22** to move axially.

(58) It may be understood that the third sleeve pipe **363** is always disposed to pass through the third through hole **3314** during the movement of the second sliding block **331**, such that the third sleeve pipe **363** is utilized to exert a better axial guiding effect on the control fiber **23**, thus preventing the control fiber **23** from being pressed and bent.

(59) In some embodiments, a clamping structure is disposed at the sliding fit portion between the first sliding block **321** and the linear guide rail **380**, and/or, the sliding fit portion between the second sliding block **331** and the linear guide rail **380**, such that the clamping structure may exert a better effect of prompting being in place, thereby making the manipulation simpler when the first sliding block **321** or the second sliding block **331** slides to the corresponding position.

(60) A clamping structure between the first sliding block **321** and the linear guide rail **380** is merely set as an example for description.

(61) In combination with FIG. 8, one or more arc-shaped grooves **381** are disposed on the side of the linear guide rail **380**; an elastic member **323** and a ball **324** are disposed in the first sliding block **321**; the ball **324** is elastically abutted on one side of the linear guide rail **380** provided with the arc-shaped groove **381** under the action of the elastic member **323**. When the first sliding block **321** moves along the linear guide rail **380** until the ball **324** is opposite to the arc-shaped groove **381**, partial structures of the ball **324** are abutted into the arc-shaped groove **381**. Therefore, a sound is generated due to the impact between the ball **324** and the linear guide rail **380** at the arc-shaped groove **381** once the ball **324** is abutted into the arc-shaped groove **381**, thus exerting a

better effect of prompting being in place, achieving simple operation and avoiding poor effect caused by insufficient manipulation or over-manipulation. In addition, the opening on the housing may exert the limiting effect in axial length, and may prompt the maximum movement limit for the operator better in coordination with the sound feedback effect of the elastic member and the ball. (62) Accordingly, the above clamping structure may be configured by the second sliding block **331** and the linear guide rail **380** directly, so as to prompt the second sliding block **331** being positioned in place, which will not be described in detail herein.

(63) It may be understood that the number and position of the arc-shaped grooves **381** on the linear guide rail **380** may be configured according to the positioning requirements of the first sliding block **321** and/or the second sliding block **331**. For example, in some embodiments, as shown in FIG. **8**, two arc-shaped grooves **381** are respectively disposed at the proximal end **380a** and the distal end **380b** of the linear guide rail **380** in the extension direction thereof; and an arc-shaped groove **381** is disposed in the middle, such that when the first sliding block **321** moves towards the distal end **380a** along the linear guide rail **380** to be opposite to the arc-shaped groove **381** at the distal end **380a**. Since the ball **324** disposed in the first sliding block **321** will make an impact sound when the ball **324** is clamped into the arc-shaped groove **381** under the elastic force of the elastic member **323**, which may exert a prompt effect when the first sliding block **321** slides to the distal end **380a** of the linear guide rail **380**. Accordingly, when the second sliding block **331** moves toward the proximal end **380b** along the linear guide rail **380** to be opposite to the arc-shaped groove **381** at the proximal end, the ball **324** disposed in the second sliding block **331** is clamped into the arc-shaped groove **381** located at the proximal end **380b**. The above configuration also may exert a prompt effect when the second sliding block **331** slides relative to the linear guide rail **380**.

(64) The elastic member **323** may be a spring, or may be an elastic bar or rod made of an elastic material.

(65) The prompt of being in place may be achieved between the first sliding block **321**/the second sliding block **331** and the linear guide rail **380** by a same structural form.

(66) An assembly structure between the first sliding block **321** and the linear guide rail **380** is merely set as an example to further describe the mounting forms of the elastic member **323** and the ball **324**.

(67) With continuing reference to FIG. **8**, a through hole **321a** is disposed on the side wall of the first sliding block **321**; the through hole **321a** penetrates to the side wall of a sliding hole **30a** of the first sliding block **321**; and the elastic member **323** is abutted against and communicated with the ball **324** to be mounted into the through hole **321a** together, such that the ball **324** is abutted to the linear guide rail **380** in sliding fit with the sliding hole **30a**. A stopping member, such as, a spacer **325** or a baffle, is disposed on a side wall of the first sliding block **321** opened with the through hole **321a**, thus positioning the elastic member **323**, thereby preventing the elastic member **323** from being withdrawn from the through hole **321a**.

(68) A spacer **325** is taken as a stopping member as an example, and the spacer **325** is locked to a mounting hole location **321b** on the first sliding block **321** via connecting members **326**, such as a screw or a bolt. The spacer **325** covers the through hole **321a** and is abutted against the elastic member **323**. Then, the elastic member **323** is in a compressed state and the ball **324** is elastically abutted against the linear guide rail **380**. Further, when the first sliding block **321** moves relative to the linear guide rail **380** such that the ball **324** is opposite to the arc-shaped groove **381**, the ball **324** is abutted into the arc-shaped groove **381** under the elastic force of the elastic member **323** to make a sound, thus prompting that the first sliding block **321** slides in position along the linear guide rail **380**. That is, the push rod **13** is driven by the first sliding block **321** to manipulate the opening or closure of the grasping portion **12**.

(69) In other embodiments, the stopping member may be also fixed on the side wall of the first sliding block **321** which is opened with the through hole **321a** by welding or gluing, and abutted against an end portion of the elastic member **323** close to the outside of the through hole **321a**, thus

limiting the elastic member **323** to be popped out from the through hole **321a**.

(70) In combination with FIGS. **3** to **5**, a conductive sliding rail **390** is fixedly disposed in the housing **31**, and the proximal end of the conductive sliding rail **390** is connected to a conductive interface (not shown) at a proximal end portion of the housing. The third manipulation portion **34** is slidably connected to the conductive sliding rail **390**, i. e., the third manipulation portion **34** may move axially relative to the conductive sliding rail **390**. A conductive elastic piece **343** is disposed at a sliding connection portion between the third operation manipulation portion **34** and the conductive sliding rail **390**; the conductive elastic piece **343** is fixedly connected to the third manipulation portion **34**; and when the third manipulation portion **34** slides along the conductive sliding rail **390**, the conductive elastic piece **343** keeps elastically abutted against the conductive sliding rail **390**. For example, the third manipulation portion **34** includes a third sliding block **341** and a third pushing member **342**. The third sliding block **341** is slidably disposed on the conductive sliding rail **390** and connected to the proximal end **23a** of the control fiber **23**, and a third pushing member **342** is connected to the third sliding block **341**. Partial structures of the third pushing member **342** are exposed to the housing **31** so as to manipulate the third pushing member **342**; the third pushing member **342** may move axially relative to the housing **31**, such that the third sliding block **341** drives the control fiber **23** to move axially; a conductive elastic piece **343** is disposed at the sliding connection portion of the third sliding block **341** and the conductive sliding rail **390**, and the conductive elastic piece is fixedly connected to the third sliding block. When the third manipulation portion **34** slides along the conductive sliding rail **390**, the conductive elastic piece **343** keeps abutted against the conductive sliding rail **390**, such that the conductive elastic piece **343** always maintains electric contact with the conductive sliding rail **390**, thus meeting the power supply requirements of the cutting device **20**.

(71) In combination with FIG. **9**, the third sliding block **341** has a cuboid structure. In other embodiments, the third sliding block **341** may be in other shapes, for example, the third sliding block **341** is in a triangular or elliptical shape in the cross section perpendicular to an axial direction. The shape of the third sliding block **341** is not defined here as long as the third sliding block **341** may move along the conductive sliding rail **390**, thus driving the control fiber **23** to move axially in the catheter body **11**.

(72) The third sliding block **341** has an axially extending through hole **341a** having an inner diameter slightly larger than an outer diameter of the second telescopic sleeve pipe **372**, such that the second telescopic sleeve pipe **372** is inserted into the through hole **341a**. The third sliding block **341** further has a first threaded hole **341b** and a second threaded hole **341c**.

(73) The first threaded hole **341b** is configured for cooperating with connecting members, such as, a screw or a bolt, such that the third sliding block **341** is fixedly connected with the third pushing member **342**. The number and the position of the first threaded holes **341b** on the third sliding block **341** may be reasonably configured according to the connection requirements of the third sliding block **341** and the third pushing member **342**, which is not defined here.

(74) The second threaded hole **341c** is orthogonal to the position of the through hole **341a**. After the proximal end of the second telescopic sleeve pipe **372** and the proximal end of the control fiber **23** located therein are inserted into the through hole **341a**, fasteners, such as, a screw or a top thread are threadedly engaged at the second threaded hole and abutted against the proximal end of the second telescopic sleeve pipe **372** and the proximal end **23a** of the control fiber **23**, such that the proximal end of the second telescopic sleeve pipe **372** and the proximal end of the control fiber **23** are fixed connected with the third sliding block **341**.

(75) It may be understood that, in some embodiments, the second telescopic sleeve pipe **372** may be omitted, namely, the control fiber **23** is directly disposed to pass through the third sleeve pipe **363**. At this time, the fastener is threadedly engaged at the second threaded hole **341c**, abutted against the proximal end **23a** of the control fiber **23** inserted into the through hole **341a**, such that the control fiber **23** is connected with the third sliding block **341**. Furthermore, the third sliding

block **341** may manipulate the control fiber **23** to move axially in the catheter body **11** under the action of the third pushing member **342**.

(76) With continuing reference to FIG. **9**, the third sliding block **341** further has a square through hole **341d** having a size much greater than the size of the cross section of the conductive sliding rail **390**, such that the third sliding block **341** has sufficient mounting space to arrange the conductive elastic piece **343** when the third sliding block **341** is slidingly engaged with the conductive sliding rail **390** through the square through hole **341d**.

(77) For example, a mounting hole **341e** is opened on a side wall of the square through hole **341d** on the third sliding block **341**; the conductive elastic piece **343** has a connecting portion **3431** provided with a through hole **343a**, and the conductive elastic piece **343** is mounted into the square through hole **341d**. Connecting members, such as a screw or bolt, pass through the through hole **343a** of the conductive elastic piece **343** and are fixed on the mounting hole **341e**, such that the conductive elastic piece **343** is fixedly connected with the third sliding block **341**. It may also be understood that in this embodiment, since the control fiber **23** is not in direct contact with the conductive elastic piece **343** or the conductive sliding rail **390**, the third sliding block **341** should be made of a conductive material. It can be appreciated that in other embodiments, when the control fiber is directly connected to a conductive elastic piece or a conductive sliding rail, or indirectly connected via other electric conductors, the third sliding block is no longer limited to being made of a conductive material.

(78) In other embodiments, the conductive elastic piece **343** may also be fixed at the square through hole **341d** of the third sliding block **341** by welding, glue bonding or buckle connection, or the like only when the conductive sliding rail **390** is in sliding fit with the third sliding block **341**, the conductive elastic piece **343** is in elastic contact with the conductive sliding rail **390**, thus achieving an electric connection therebetween.

(79) The conductive elastic piece **343** has one or two elastic abutting portions **3432** bent and projected by the connecting portion **3431**, such that when the conductive sliding rail **390** is assembled into the square through hole **341d** of the third sliding block **341**, the elastic abutting portions **3432** are elastically abutted against and keep an electric contact with the conductive sliding rail **390**. Therefore, when the third sliding block **341** moves along the conductive sliding rail **390**, the conductive elastic piece **343** still keeps a good electric contact with the conductive sliding rail **390**, so as to meet the power supply requirements of the cutting device **20**.

(80) The elastic abutting portion **3432** is “V”-shaped or “U”-shaped so as to maintain a line contact or a surface contact with the conductive sliding rail **390**, thus obtaining a better electric contact effect. Meanwhile, the above configuration reduces the sliding friction of the conductive elastic piece **343** during the movement of the third sliding block **341** relative to the conductive sliding rail **390** so as to avoid abrasion, thereby ensuring stable electric contact between the conductive elastic piece **343** and the conductive sliding rail **390**.

(81) In other embodiments, the structure where the third sliding block **341** is in sliding fit with the conductive sliding rail **390** may not be a square through hole **341**. For example, if the conductive sliding rail **390** is an I-shaped guide rail, the structure where the third sliding block **341** is in fit connection with the conductive sliding rail **390** may be a sliding groove with an I-shaped cross section. For another example, if the conductive sliding rail **390** has a triangular cross section, the sliding groove where the third sliding block **341** is in fit connection with the conductive sliding rail **390** has a triangular cross section.

(82) It may be understood that the conductive elastic piece **343** and the conductive sliding rail **390** may be made of easily conductive metal materials such as, iron, aluminum, copper. But the material of the conductive elastic piece **343** is not limited herein as long as the conductive elastic piece **343** may be in electric contact with the conductive sliding rail **390** so as to meet the power supply requirements of the cutting device **20**.

(83) A conductive structure is disposed in the housing, and when the conductive interface at the

proximal end of the housing is powered on, it may be ensured that the third sliding block is also always in a powered state, thus avoiding the failure of cutting caused by the invalid electric connection when cutting tissues.

(84) It can be appreciated that in other embodiments, the electric connection between the conductive sliding rail and the third sliding block may also be implemented by other conductive structures as long as it is ensured that the third sliding block is always electrically connected to the conductive structure during sliding process. For example, an electric brush may be disposed on the third sliding block, and the electric brush is always in contact with the surface of the conductive sliding rail; alternatively, the conductive structure may include a spring; the spring may be disposed at a position where the third sliding block is electrically connected to the conductive sliding rail; one end of the spring is fixedly connected to the third sliding block and always in a compressed state.

(85) It can also be appreciated that in other embodiments, the conductive sliding rail may not be provided, i. e., the third sliding block is directly connected to the conductive interface at the proximal end of the housing via the conductive structure.

(86) In the embodiments, the configuration of a limiting structure for limiting the elongated member to prevent the elongated member from buckling, an opening on the housing for limiting the axial movement range of the manipulation portion, and an elastic member and a ball for prompting the movement in place, a conductive structure for ensuring that the third manipulation portion is always powered on during the movement is to ensure that the neostomy apparatus is valid in each operation process of the neostomy apparatus in use, such that the grasping device may successfully grasp the tissue, and the cutting device may also successfully cut the tissue.

(87) Each feature of the above embodiments may be in any combination. To make the description concise, all the possible combinations of each feature in the above embodiments are not described one by one. However, any combination of these features shall fall within the scope of the embodiments as long as there is no contradiction therebetween.

(88) The above examples are merely illustrative of several embodiments of the embodiments, and are described in greater detail, but are not construed as limiting the embodiments. It is understood that any person skilled in the art may further make several variations and modifications within the concept of the embodiments, and these variations and modifications fall within the scope of the embodiments.

Claims

1. A neostomy apparatus, comprising: a grasping device, a cutting device and a control handle; the control handle comprises a housing and a manipulation portion disposed in the housing, the manipulation portion is configured to axially move relative to the housing; the grasping device or the cutting device comprises an elongated member, and the manipulation portion drives the elongated member to move axially; the elongated member is partially located in the housing, and a first limiting structure is further disposed in the housing, the first limiting structure has a groove or a through hole, and a portion of the elongated member located in the housing is at least partially disposed in the groove or the through hole of the first limiting structure, the elongated member is configured to axially move relative to the first limiting structure; and the elongated member resists bending due to the radial constraint of the first limiting structure; wherein the grasping device further comprises a catheter body, the elongated member passing through the catheter body; wherein the elongated member comprises a push rod, a traction fiber, and a control fiber; wherein the manipulation portion comprises a first manipulation portion, a second manipulation portion, and a third manipulation portion; wherein the first manipulation portion, the second manipulation portion, the third manipulation portion are configured to manipulate an axial movement of the push rod, the traction fiber, and the control fiber in the catheter body, respectively; wherein the cutting

device comprises a cutting portion, the cutting portion being connected to distal ends of the traction fiber and the control fiber; and wherein the traction fiber and the control fiber are symmetrically disposed on a wall of the catheter body to exert opposite axial push-pull forces on the cutting portion.

2. The neostomy apparatus according to claim 1, wherein the first limiting structure comprises a first sleeve pipe, a second sleeve pipe, and a third sleeve pipe, which correspond to the first manipulation portion, the second manipulation portion and the third manipulation portion, respectively; wherein a proximal end of each of the first sleeve pipe, the second sleeve pipe, and the third sleeve pipe is close to a distal end of the corresponding first manipulation portion, second manipulation portion, and third manipulation portion, respectively.

3. The neostomy apparatus according to claim 1, wherein a proximal end of the catheter body is connected to the housing; the housing is provided with a sealing assembly; the sealing assembly is located at a distal end of the housing and is sealingly connected to the proximal end of the catheter body, and the catheter body is configured to be communicated with a negative pressure source through the sealing assembly, such that a negative pressure is formed in the catheter body under the action of the negative pressure source.

4. The neostomy apparatus according to claim 3, wherein the sealing assembly comprises a hollow pipe body and a sealing member connected at both ends of the hollow pipe body; a vent hole is opened on a side wall of the hollow pipe body, and the negative pressure source is communicated with the vent hole via a hose.

5. The neostomy apparatus according to claim 1, wherein, the second manipulation portion or the third manipulation portion is connected with a second limiting structure; the second limiting structure is configured to receive the elongated member and is coaxial with the first limiting structure; a proximal end of the second limiting structure and a proximal end of the elongated member are fixed on the second manipulation portion or the third manipulation portion; a distal end of the second limiting structure is inserted into the groove or the through hole of the first limiting structure and is configured to move axially relative to the first limiting structure under the action of the second manipulation portion or the third manipulation portion.

6. The neostomy apparatus according to claim 1, wherein a linear guide rail is fixedly disposed in the housing; the first manipulation portion and the second manipulation portion are slidably disposed on the linear guide rail; one or more arc-shaped grooves are disposed on a side of the linear guide rail; each of the first manipulation portion and the second manipulation portion comprises an elastic member and a ball; the ball is elastically abutted against one side of the linear guide rail opened with the arc-shaped groove under the action of the elastic member; and, when each of the first manipulation portion and the second moves along the linear guide rail until the ball is opposite to the arc-shaped groove, partial structures of the ball are abutted against the arc-shaped groove.

7. The neostomy apparatus according to claim 1, further comprising a conductive structure and a conductive interface disposed at a proximal end portion of the housing; the conductive structure always keeps electric connection with the third manipulation portion and the conductive interface when the third manipulation portion moves axially.

8. The neostomy apparatus according to claim 7, wherein the housing is provided with a conductive sliding rail, and a proximal end of the conductive sliding rail is connected to the conductive structure; and the third manipulation portion is slidably connected to the conductive sliding rail, and the third manipulation portion is configured to move axially relative to the conductive sliding rail.

9. The neostomy apparatus according to claim 8, wherein the conductive structure comprises a conductive elastic piece, and the conductive elastic piece is disposed at a sliding connection portion between the third manipulation portion and the conductive sliding rail; when the third manipulation portion moves axially along the conductive sliding rail, the conductive elastic piece is always

abutted against the conductive sliding rail.

10. The neostomy apparatus according to claim 1, wherein each of the first manipulation portion, the second manipulation portion, and the third manipulation portion comprises a sliding block and a pushing member, and the sliding block is slidably disposed in the housing and is connected to the proximal end of the elongated member; the pushing member is connected to the sliding block, and partial structures of the pushing member are exposed to the housing, and the pushing member is configured to move axially relative to the housing, such that the sliding block drives the elongated member to move axially.

11. The neostomy apparatus according to claim 1, wherein the cutting device comprises a cutting portion, the cutting portion being connected to distal ends of the traction fiber and the control fiber; and wherein the traction fiber and the control fiber are symmetrically disposed on a wall of the catheter body to exert opposite axial push-pull forces on the cutting portion.

12. The neostomy apparatus according to claim 11, wherein the first manipulation portion is connected to the proximal end of the push rod and is configured for driving the push rod to move axially relative to the catheter body.

13. The neostomy apparatus according to claim 12, wherein the second manipulation portion is connected to the proximal end of the traction fiber and is configured for driving the traction fiber to move axially in the catheter body.

14. The neostomy apparatus according to claim 13, the third manipulation portion is connected with the proximal end of the control fiber and is configured for driving the control fiber to move axially in the catheter body.

15. The neostomy apparatus according to claim 2, wherein the push rod is configured to axially move relative to the first sleeve pipe; and wherein the traction fiber is configured to axially move relative to the second sleeve pipe, and wherein the control fiber is configured to axially move relative to the third sleeve pipe.

16. The neostomy apparatus according to claim 2, wherein the first manipulation portion comprises a first sliding block and a first pushing member, and the first sliding block is slidably disposed in the housing and is connected to the proximal end of the push rod; the first pushing member is connected to the first sliding block, and partial structures of the first pushing member are exposed to the housing; and the first pushing member is configured to move axially relative to the housing, such that the first sliding block drives the push rod to move axially.

17. The neostomy apparatus according to claim 16, wherein the second manipulation portion and the third manipulation portion are both located on one side where the proximal end of the first manipulation portion is located; the first sliding block is provided with a first through hole and a second through hole which extend in an axial direction; wherein the second sleeve pipe and the third sleeve pipe are axially and movably disposed to pass through the first through hole and the second through hole, respectively.

18. The neostomy apparatus according to claim 16, wherein the second manipulation portion comprises a second sliding block and a second pushing member, and the second sliding block is slidably disposed in the housing and is connected to the proximal end of the traction fiber; the second pushing member is connected to the second sliding block, and partial structures of the second pushing member are exposed to the housing, and the second pushing member is configured to move axially relative to the housing, such that the traction fiber is driven by the second sliding block to move axially.

19. The neostomy apparatus according to claim 18, wherein the second sliding block comprises a third through hole, and the third sleeve pipe is configured to move axially relative to the third through hole.

20. The neostomy apparatus according to claim 18, wherein the housing is provided with a conductive sliding rail, and the proximal end of the conductive sliding rail is connected to the conductive interface at a proximal end portion of the housing, the third manipulation portion is

slidably connected to the conductive sliding rail, and the third manipulation portion is configured to move axially relative to the conductive sliding rail; wherein the third manipulation portion comprises a third sliding block and a third pushing member, and the third sliding block is slidably disposed on the conductive sliding rail and connected to the proximal end of the control fiber; the third pushing member is connected to the third sliding block, and partial structures of the third pushing member are exposed to the housing; the third pushing member is configured to move axially relative to the housing, such that the third sliding block drives the control fiber to move axially; wherein a conductive elastic piece is disposed at the sliding connection portion of the third sliding block and the conductive sliding rail, and the conductive elastic piece is fixedly connected to the third sliding block.
